PALAWAN STATE UNIVERSITY
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Bachelor of Science in Business Administration
major in Marketing Management

The Impact of Artificial Intelligence on Employment Opportunities for Marketing Management Graduates

Researchers

SADANG, LOREN
MADRIAGA, CLEHIAH U.
FABRIGAS, PRINCESS AYAH
CASERES, ROSELYN
TORINO, ANGEL MAE D.
OBSANGA, JANNA A.
CABALLES, JUNELLE JOSH
ASIRON, SITTI KARIJA H.
ORDAS, REYZEL JOYCE
HANSEN, RICO V.
OBEN, SHERILYN
VIRAY, SUNSHINE

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CHAPTER 1

INTRODUCTION

Background of the Study

Generative artificial intelligence (AI) refers to computational systems that produce text, images, code, and other content from patterns learned from large datasets. Its development is grounded in natural language processing and the transformer architecture, which allowed models to weigh relationships among words in parallel and to be trained at far larger scales (Vaswani et al., 2017). Brown et al. (2020) subsequently demonstrated that scaling transformer-based language models produced broad, few-shot capabilities, providing the technical foundation for general-purpose large language models (LLMs). On November 30, 2022, OpenAI released ChatGPT, a conversational interface built on its GPT models (OpenAI, 2022). The release made generative AI directly accessible to users without specialized technical training and shifted LLMs from research systems toward everyday productivity tools.

Capabilities expanded rapidly after ChatGPT's release. GPT-4 demonstrated stronger performance across professional and academic examinations and supported image inputs alongside text (OpenAI, 2023). Subsequent systems improved reasoning, coding, document analysis, and multimodal interaction, while competition and investment lowered cost and expanded access. Continued gains are expected as model scale, training data, instruction tuning, and tool use improve. These advances have moved generative AI into tasks once treated as distinctly human, including drafting, summarizing, coding, analysis, and customer interaction.

The expected labor-market consequences have become a major policy concern. The World Economic Forum (2025) projects that, by 2030, 170 million jobs will be created and 92 million displaced, producing a net gain of 78 million jobs. The same report estimates that 39% of key skills will change and identifies AI and information processing as technologies associated with both substantial job creation and displacement. These projections indicate that AI is unlikely to affect work uniformly: some occupations may expand, others may contract, and many are expected to be reorganized around new task combinations.

Recent evidence suggests that business and finance occupations are especially exposed. Massenkoff and McCrory (2026) distinguish theoretical coverage, or the share of an occupation's tasks that current LLMs can perform, from observed exposure, which also considers actual use. Business and financial work ranked high in theoretical coverage, and roles such as market research analysts and marketing specialists were among the most exposed. Marketing work combines writing, market research, data interpretation, campaign analysis, and client communication, making it directly relevant to these findings. Exposure, however, does not automatically mean replacement; AI may automate discrete tasks while augmenting judgment, strategy, and relationship-building.

Because marketing graduates are likely to encounter AI in the workplace, their readiness depends partly on what they know about it. Familiarity is uneven among students and the wider public. In a survey of higher-education students, Chan and Hu (2023) found varied familiarity with generative AI and a mixture of perceived benefits and concerns, including accuracy and academic integrity. AI literacy is also multidimensional, involving conceptual understanding, evaluation, creation, and ethical use rather than mere access to tools (Ng et al., 2021). Differences in exposure, discipline, access, and instruction can therefore produce uneven confidence and competence.

Within the Philippines, AI adoption has gained policy attention but remains uneven. The Philippine AI Roadmap identifies business productivity, innovation, and workforce readiness as priorities and encourages AI use across key sectors (Department of Trade and Industry, 2021). For marketing graduates, this policy direction means entering a labor market where employers may differ in their access to and use of AI. Larger firms and those in highly digitalized industries may adopt specialized tools sooner, while smaller enterprises may depend on low-cost, general-purpose applications. Such uneven adoption can widen existing differences in graduate readiness and opportunity.

Measuring that readiness is itself a challenge. Self-assessments of skill are often inaccurate, and people with lower skill tend to overestimate their own performance (Kruger & Dunning, 1999). A graduate's own rating of AI competence may therefore reflect confidence more than ability. In addition, a graduate's time to find suitable work depends not only on AI

skills but also on how strong a candidate the graduate is and how actively the graduate searches. A study of AI skills and employment opportunities should thus measure proficiency through performance rather than self-report and should account for these other influences.

Taken together, these developments make AI's relationship to Marketing Management graduates' employment opportunities a relevant inquiry, especially as business, finance, and marketing tasks show high exposure and students' AI knowledge remains uneven.

Statement of the Problem

AI is increasingly entering marketing work through content generation, market research, campaign analysis, and customer engagement. For Marketing Management graduates, this shift raises questions about employer requirements, hiring prospects, and the kinds of opportunities available to them. The study therefore examines whether AI skills are becoming a hiring requirement, whether graduates with demonstrated AI proficiency gain greater access to employment after accounting for how strong a candidate they are and how actively they search, how AI is changing their opportunities, and how they perceive those opportunities. Specifically, the study seeks to answer the following questions:

1. To what extent are AI skills increasingly required by employers for marketing-related positions?
   a. Which AI-related competencies are most frequently required?
   b. How does demand differ across marketing roles and industries?
2. Is there a significant relationship between AI skills, as measured by a performance-based proficiency test, and employment opportunities for Marketing Management graduates, after accounting for candidate strength and job search intensity?
   a. To what extent do AI skills influence employers' hiring decisions?
   b. Which employment opportunities (time to employment, interview and offer rates, and degree relevance) are associated with AI skills?
3. How has AI affected employment opportunities for Marketing Management graduates?
   a. Which marketing tasks are being automated, augmented, or newly created?

b. How have entry-level requirements and hiring criteria changed?

4. How do Marketing Management graduates perceive their employment opportunities in an AI-driven labor market?

a. How prepared do they feel to use AI in professional marketing work?

b. What opportunities and threats do they associate with AI?

Hypotheses

The following null and alternative hypotheses correspond to Research Questions 1 through 4 and are tested at a 0.05 level of significance. A null hypothesis is rejected when $p < .05$. The alternative hypothesis for Research Question 1 is directional (an increase), and the others are non-directional.

Research Question 1

$H0_1$: Employer demand for AI skills in marketing-related positions has not significantly increased.

$H1_1$: Employer demand for AI skills in marketing-related positions has significantly increased.

Research Question 2

$H0_2$: After controlling for candidate strength and job search intensity, there is no significant relationship between AI skills and the employment opportunities of Marketing Management graduates.

$H1_2$: After controlling for candidate strength and job search intensity, there is a significant relationship between AI skills and the employment opportunities of Marketing Management graduates.

Research Question 3

$H0_3$: AI has had no significant effect on the employment opportunities of Marketing Management graduates through changes in marketing tasks and hiring criteria.

$H1_3$: AI has had a significant effect on the employment opportunities of Marketing Management graduates through changes in marketing tasks and hiring criteria.

Research Question 4

$H0_4$: Marketing Management graduates do not perceive AI as presenting significant opportunities or threats to their employment, and their reported preparedness does not vary.

$H1_4$: Marketing Management graduates perceive AI as presenting both significant opportunities and threats to their employment, and their reported preparedness varies.

Conceptual Framework

The conceptual framework of the study draws on four ideas from the reviewed literature. First, the exposure of an occupation to AI depends on the share of its tasks that AI can perform, and exposure does not by itself imply replacement (Massenkoff & McCrory, 2026). Second, the skills that employers require are expected to change as AI and information processing technologies spread (World Economic Forum, 2025). Third, a graduate's readiness depends on AI literacy, which covers understanding, evaluation, creation, and ethical use rather than access to tools alone (Ng et al., 2021). Fourth, because self-assessments of skill are unreliable (Kruger & Dunning, 1999), AI proficiency is measured through a scored performance test, and because job search outcomes also depend on candidate strength and search effort, these two factors are treated as control variables. Together, these ideas explain why employer requirements, graduates' skills, and graduates' perceptions are examined in one study.

Figure 1 presents the input-process-output paradigm of the study. The inputs are job posting data for marketing-related positions and survey responses from Marketing Management graduates, including their scores on an AI proficiency test. These are processed through content analysis, survey administration, and statistical treatment. The output is a set of findings on employer demand for AI skills, the effect of AI on hiring and employment opportunities, and graduates' preparedness and perceptions.

Figure 1

Input-Process-Output Paradigm of the Study

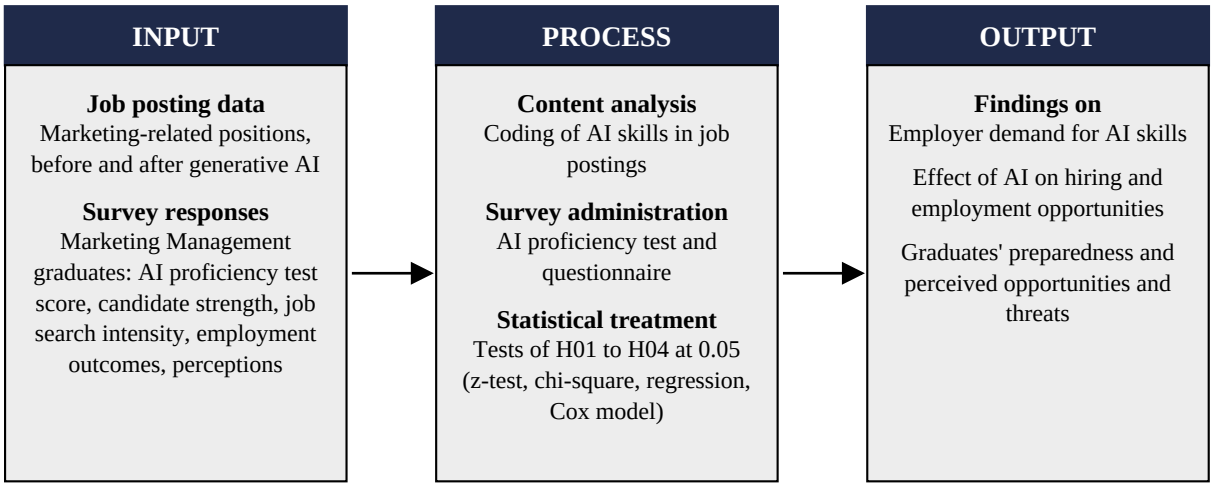
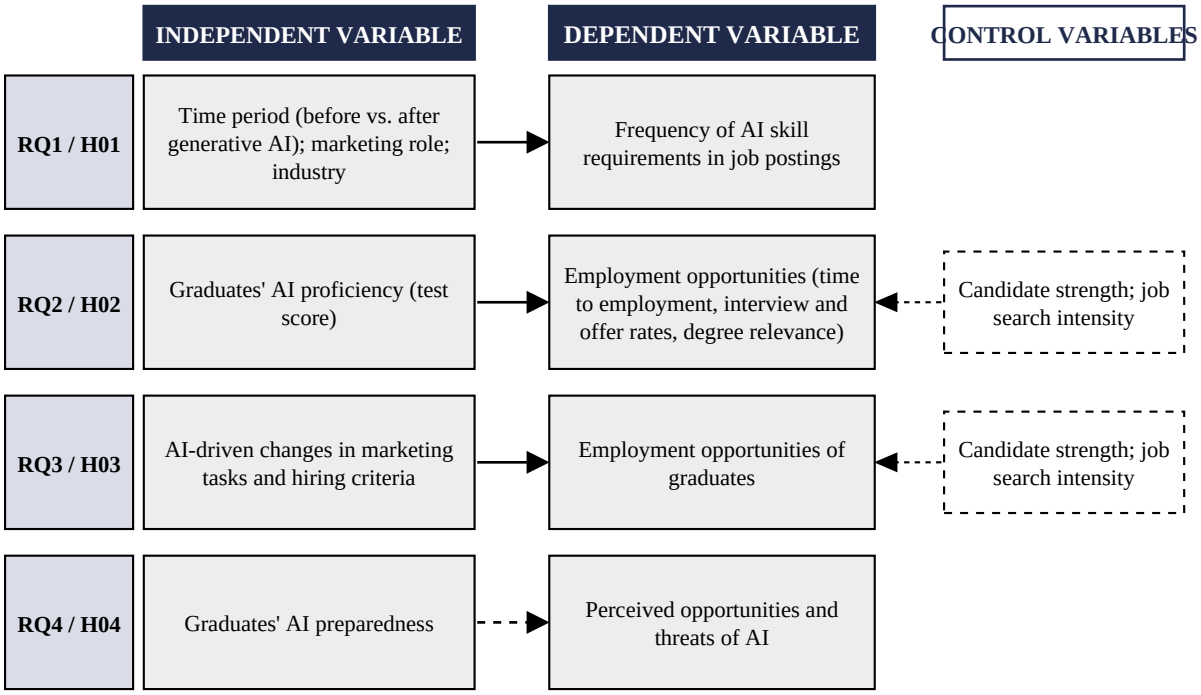


Figure 2 identifies the independent, dependent, and control variables associated with each research question and hypothesis. For Research Question 1, the time period, the marketing role, and the industry are treated as independent variables, and the frequency with which AI skills are required in job postings is the dependent variable. For Research Questions 2 and 3, the employment opportunities of Marketing Management graduates are the dependent variable, influenced by graduates' AI proficiency and by AI-driven changes in marketing tasks and hiring criteria, respectively. In both, candidate strength and job search intensity are held constant as control variables. Research Question 4 is largely descriptive, so its relationship is shown with a dashed arrow: graduates' AI preparedness is examined alongside their perceived opportunities and threats.

Figure 2

Independent, Dependent, and Control Variables by Research Question



Note. Solid arrows indicate relationships tested inferentially. The dashed arrow indicates a descriptive relationship. Dotted lines from the control variables indicate factors held constant in the analysis.

Significance of the Study

This study examines how artificial intelligence is changing employment opportunities for Marketing Management graduates in the Philippines, including opportunities in the international labor market. Its findings may help identify the competencies graduates need to remain qualified and competitive as marketing work, employer expectations, and hiring practices continue to change. The study may be significant to the following:

Marketing Management Graduates. The findings may help graduates understand how AI is influencing marketing-related employment, the competencies employers increasingly seek, and the ways AI is changing marketing tasks and hiring criteria. By identifying relevant AI skills and areas of preparedness, the study may help graduates make informed decisions about developing their abilities and adapting to changes in the profession. It may also provide insight into opportunities in both the Philippine and international labor markets.

Marketing Management Students. The study may help current students recognize the importance of developing AI literacy and practical AI-related competencies before entering the workforce. Its findings may encourage them to prepare for changes in marketing roles rather than relying solely on the knowledge and skills traditionally associated with the degree.

Educational Institutions and Marketing Management Educators. The findings may provide information that can be considered when reviewing Marketing Management curricula, learning activities, and career-preparation programs. In particular, they may help identify areas where instruction on AI tools, critical evaluation, ethical use, and the application of AI in marketing could be strengthened.

Employers and Businesses. The study may offer insight into the AI-related competencies of prospective Marketing Management employees and the changing requirements of marketing positions. This may help employers consider how AI can support marketing work while retaining the human judgment, creativity, and professional skills required in their organizations.

Career Guidance Offices and Employment Support Organizations. The findings may help these groups provide more relevant career guidance by identifying changes in marketing job requirements and the skills graduates may need when seeking suitable employment in the Philippines or abroad.

Future Researchers. This study may serve as a reference for future research on AI adoption, AI literacy, labor-market changes, and employment opportunities in marketing and related business fields. It may also provide a basis for studies that examine these developments across other locations, industries, or graduate populations.

Scope and Delimitation

This study focuses on the influence of artificial intelligence, particularly generative AI systems based on large language models (LLMs), on employment opportunities for Marketing Management graduates in the Philippines. It examines changes in employer demand for AI-related skills, the relationship between graduates' AI competencies and their access to marketing-related employment, changes in marketing tasks and hiring criteria, and graduates'

perceptions of opportunities, threats, and preparedness in an AI-driven labor market.

The study considers employment opportunities in the Philippines as well as relevant opportunities available to Filipino Marketing Management graduates in the international labor market. Overseas opportunities are considered as an additional dimension of the employment landscape; the study does not attempt to provide a separate, exhaustive analysis of every foreign labor market, country, or immigration system.

The study's respondents are limited to Filipino Marketing Management graduates who graduated in 2023 or later, so that their job search took place after the public release of ChatGPT, and who are currently working in marketing-related positions aligned with Marketing Management, or who are actively seeking employment in such positions. The focus is on opportunities within the field of Marketing Management rather than employment in unrelated occupations or positions that do not make meaningful use of marketing-related qualifications. Accordingly, graduates who are employed only in unrelated work are not treated as evidence of employment in a Marketing Management position.

To examine changes in employer requirements, the study also considers job postings for marketing-related positions from periods before and after the public release of ChatGPT on November 30, 2022. The comparison is intended to identify changes in the occurrence and nature of AI-related skill requirements. The analysis is limited to the job postings that can be collected and that meet the study's selection criteria; therefore, the results may not represent every vacancy or employer in the Philippines or overseas.

The study is limited to employment-related effects and perceptions. It does not examine the technical development, training, or internal architecture of AI models, nor does it evaluate every form of artificial intelligence. Although AI is discussed broadly for conceptual purposes, the study gives particular attention to generative AI based on LLMs and their use in tasks relevant to marketing. It does not assume that AI exposure necessarily results in the complete replacement of marketing employees. Rather, it considers the possible automation of particular tasks, the augmentation of human work, and the emergence or alteration of job requirements.

AI proficiency is measured through a scored test of knowledge and applied skill, not through self-rating, and the test is limited to general-purpose generative AI use in marketing tasks rather than any specific commercial tool. The findings will also be subject to the availability, accuracy, and coverage of the collected job postings and survey responses. Because the study relies on these sources, its results should be interpreted within the defined respondent group, selected job postings, and study period, rather than generalized to all graduates or all occupations.

Definition of Terms

The following terms are defined conceptually and operationally to clarify how they are used in this study:

Artificial Intelligence (AI). Artificial intelligence refers broadly to computer-based systems designed to perform tasks commonly associated with human cognitive abilities, such as learning, reasoning, problem-solving, understanding language, recognizing patterns, and making recommendations or decisions. This includes generative AI systems capable of producing content and carrying out various knowledge-based tasks. In this study, AI is discussed broadly as the technology influencing marketing work, while the specific systems of primary interest are generative AI systems based on large language models (LLMs).

Generative Artificial Intelligence. Generative artificial intelligence refers to AI systems that generate new content, such as text, images, audio, video, or computer code, based on patterns learned from data. In this study, the term refers particularly to generative systems used to support or perform tasks relevant to marketing, including content creation, research, analysis, and communication.

Large Language Models (LLMs). Large language models are AI models trained on large amounts of text and other language data to process and generate human-like language. They can perform tasks such as drafting, summarizing, answering questions, analyzing information, and assisting with problem-solving. LLMs are a subset of generative AI and are the primary type of AI system considered in this study. Examples include the language models that power conversational AI applications such as ChatGPT, Claude, and Gemini.

AI Skills. AI skills refer to the practical abilities needed to use AI systems effectively in work-related tasks. In this study, these include selecting appropriate AI tools, writing and refining prompts, evaluating generated outputs, using AI-assisted information in marketing tasks, and recognizing the limitations of AI-generated content. In the analysis of job postings, AI skills are the AI-related competencies that employers list. In the graduate survey, AI skills are measured through a scored performance test rather than self-rating.

AI Proficiency Score. The AI Proficiency Score is the percentage of the total points a respondent earns on the AI proficiency test, which combines knowledge items, prompt evaluation items, output verification items, and applied tasks. It ranges from 0 to 100, and higher scores indicate greater demonstrated ability.

AI Literacy. AI literacy refers to the knowledge and competencies needed to understand, use, evaluate, and responsibly interact with AI systems. In this study, it concerns graduates' ability to understand the capabilities and limitations of AI, assess its outputs, apply it appropriately to marketing work, and consider ethical implications.

Employment Opportunities. Employment opportunities refer to the availability and accessibility of suitable paid work. In this study, the term is limited to marketing-related positions aligned with Marketing Management, including access to relevant vacancies, hiring processes, interviews, and employment offers in the Philippines and, where applicable, overseas. It is measured through time to employment, interview rate, offer rate, and the degree relevance of the first position.

Time to Employment. Time to employment is the number of weeks from a graduate's graduation date (or the start of active job searching, whichever is later) to the start date of the first marketing-related position. For graduates still seeking such a position, it is the number of weeks elapsed up to the date of the survey.

Candidate Strength. Candidate strength refers to the qualifications that make a graduate competitive to employers. In this study, it is an index based on general weighted average, on-the-job training or internship experience, relevant certifications, portfolio, and prior relevant work experience.

Job Search Intensity. Job search intensity refers to how actively a graduate looks for work. In this study, it is an index based on the number of applications submitted per month, the hours spent searching per week, and the number of search channels used.

Marketing Management Graduates. This refers to individuals who have completed a degree in Marketing Management. For the respondent group in this study, it specifically refers to Filipino graduates, from 2023 onward, who are currently working in Marketing Management-related positions or are actively seeking such positions.

Marketing-Related Positions. These are occupations whose principal duties involve marketing functions, such as market research, digital marketing, brand management, advertising, campaign planning, marketing analysis, and related marketing activities. In this study, positions are considered relevant when their duties align with the knowledge and competencies associated with Marketing Management, rather than merely being any job held by a Marketing Management graduate.

Automation. Automation refers to the use of technology to perform tasks with reduced or no direct human involvement. In this study, it refers to marketing tasks that AI can perform partly or fully, potentially changing the amount or type of work required from employees.

Augmentation. Augmentation refers to the use of technology to assist or enhance human work rather than fully replace the worker. In this study, it refers to the use of AI to support marketing professionals in activities such as research, content development, data interpretation, and campaign planning.

Hiring Criteria. Hiring criteria refer to the qualifications, skills, experience, and other requirements employers use when evaluating applicants. In this study, the term includes AI-related competencies and other requirements specified in marketing-related job postings.

AI Preparedness. AI preparedness refers to the extent to which an individual feels equipped to understand and use AI in a work setting. In this study, it concerns Marketing Management graduates' perceived readiness to apply AI tools, evaluate their outputs, adapt to AI-related changes in marketing work, and meet evolving employer expectations. It is measured by self-report and is distinct from the tested AI Proficiency Score.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter presents literature and studies related to the impact of artificial intelligence on employment opportunities for Marketing Management graduates. It discusses the exposure of marketing-related tasks to artificial intelligence, the changing nature of employment, employer demand for skills, the measurement of AI skills, and the opportunities and challenges associated with AI adoption. It also reviews international and Philippine findings that provide a basis for examining the employment experiences and preparedness of Marketing Management graduates.

Related Literature

Artificial Intelligence and the Changing Nature of Work

Artificial intelligence (AI) refers to computer-based systems that perform activities commonly associated with human cognitive abilities. Generative AI is a form of AI that can produce new content, including written material, images, summaries, and other outputs, based on patterns learned from data. In workplaces, these systems can assist with information processing, communication, analysis, and content development. Their importance to employment is not limited to whether a whole occupation can be automated. Rather, AI may affect particular tasks within an occupation, changing how employees perform their work and what employers expect from them.

A task-based view is useful in understanding these changes. An occupation is made up of different activities, and these activities do not all require the same abilities. Some tasks may be completed or accelerated by AI, while others continue to require human judgment, contextual understanding, creativity, accountability, and interaction with people. As a result, exposure to AI should not be treated as equivalent to the complete replacement of a worker. It may instead lead to automation of selected activities, assistance with existing responsibilities, or the creation of new duties.

AI Exposure in Marketing Management Work

Marketing Management is particularly relevant to discussions of generative AI because many marketing responsibilities involve working with information and

communication. These activities include gathering and summarizing market information, reviewing competitors, identifying consumer trends, preparing reports, drafting promotional content, organizing campaign information, and interpreting performance data. Generative AI tools can support or perform parts of these activities by producing initial drafts, summarizing documents, organizing information, suggesting content ideas, and helping users interpret data.

The exposure of marketing work can therefore be explained by the nature of its tasks rather than by the degree title alone. Market research and campaign work often require reading, writing, classification, comparison, and reporting. These are activities that large language models can assist with. For example, an AI tool may prepare a first draft of a product description, summarize customer feedback, or organize findings from a market report. A marketing professional must still check whether the information is accurate, whether the output fits the target audience, and whether the proposed message supports the organization's objectives.

Anthropic's 2026 analysis identifies market research analysts and marketing specialists as one of the occupations with high observed AI exposure. The occupation was reported at approximately 64.8% observed exposure and ranked fifth among the ten most exposed occupations in the analysis. This measure reflects the share of occupational tasks that were both considered feasible for large language models and observed in work-related AI use. It does not mean that 64.8% of marketing positions will be eliminated. Instead, it indicates that a substantial portion of tasks in the occupation overlap with activities for which AI assistance or automation is already being used (Massenkoff & McCrory, 2026).

For Marketing Management graduates, this finding is relevant because their possible work includes market research, digital marketing, advertising, brand support, campaign planning, and marketing analysis. These roles may involve tasks that are increasingly supported by AI. However, the occupation-level finding should be interpreted carefully: it is based on U.S. occupational data and Anthropic platform usage, and it does not directly measure the employment situation of Filipino graduates. It provides evidence of task exposure, while the present study examines how AI skills and changes in hiring relate to graduates' employment opportunities.

Automation, Augmentation, and Human Skills

AI can affect work through automation and augmentation. Automation occurs when technology performs a task with reduced human involvement. Augmentation occurs when technology supports employees and helps them complete work more efficiently. In marketing, examples of tasks that may be automated include producing routine text drafts, categorizing basic feedback, or preparing initial summaries. Examples of augmentation include using AI to generate alternative campaign ideas, summarize research for review, or assist with the preparation of reports.

The distinction matters because the same technology can have different effects depending on how an organization uses it. If AI is used mainly to replace routine activities, employers may need fewer workers for some task combinations. If AI is used as an assistant, employees may be able to complete more work while retaining responsibility for decisions and quality. Marketing work also involves understanding customers, interpreting a brand's identity, making strategic choices, coordinating with colleagues, and responding to changing market conditions. These responsibilities require human oversight and cannot be evaluated only by the ability of an AI tool to generate text or summarize information.

The literature therefore suggests that graduates need more than the ability to access AI tools. They need to evaluate outputs, identify errors or unsupported claims, protect confidential information, and decide when AI assistance is appropriate. Communication, critical thinking, creativity, adaptability, ethical judgment, and knowledge of marketing principles remain important because these abilities help workers apply AI outputs to real organizational goals.

Changing Employer Demand for Skills

The spread of AI changes the skills that employees may need even when their occupations do not require them to develop AI systems. Workers may need to understand how AI tools operate, select appropriate applications, give clear instructions, assess generated information, and combine AI-assisted work with their professional knowledge. Employers may also place greater value on workers who can use digital tools while maintaining sound judgment and communication.

Green (2024), in an OECD working paper, explains that most workers exposed to AI will not need specialized technical skills such as machine learning or natural language processing. Nevertheless, AI can change the tasks they perform and the skills required in their jobs. The study found that management and business skills were among the most demanded skills in occupations highly exposed to AI. It also reported an increase in vacancies requiring at least one emotional, cognitive, or digital skill. The findings indicate that AI-related workplace change is not limited to hiring technical specialists; it can also reshape the expectations placed on employees in business and other non-technical occupations.

For Marketing Management graduates, this means that employability may depend on combining marketing knowledge with practical digital and AI competencies. Knowing how to use an AI tool may be useful, but graduates must also understand consumer behavior, marketing strategy, data interpretation, communication, and ethical responsibilities. The ability to explain and defend marketing decisions remains important when AI-generated material is used as part of professional work.

AI and Employment Opportunities for New Graduates

The effects of AI on employment may differ between experienced workers and people entering the labor market. Entry-level positions often include routine research, writing, information gathering, and basic analysis. These tasks can overlap with current generative AI capabilities. If organizations use AI to perform some of these activities, they may change the number or type of entry-level positions they offer, or expect new applicants to perform more complex work from the beginning.

However, changes in hiring should not automatically be attributed to AI. Economic conditions, business decisions, interest rates, changes in work arrangements, and other factors can also affect employment. Research on AI and early-career employment therefore needs to distinguish between observed patterns and proven causal effects. This is important when interpreting international evidence and applying it to the circumstances of Marketing Management graduates in the Philippines.

Measuring AI Skills and Accounting for Other Influences on Job Outcomes

Studying how AI skills relate to employment requires a clear way of measuring those skills. Ng et al. (2021) describe AI literacy as multidimensional, covering knowing and understanding AI, using it, evaluating its outputs, and considering ethical issues. Such a range of abilities is difficult to capture by asking people to rate themselves. Research on self-assessment shows that people often misjudge their own competence, and that those with lower skill tend to overestimate their performance (Kruger & Dunning, 1999). Chan and Hu (2023) likewise found that students' familiarity with generative AI varied widely and was often accompanied by uncertainty about accuracy and appropriate use. These points suggest that a graduate's self-rating of AI skill may reflect confidence rather than ability, and that a scored performance test provides a more objective basis for comparing graduates.

Employment outcomes are also shaped by factors other than AI skills. Because time to employment depends on how competitive an applicant is and on how actively the applicant searches, any relationship between AI proficiency and job outcomes should be examined while accounting for the applicant's academic standing, practical experience, credentials, and search effort. Without these controls, a graduate who found work quickly because of strong qualifications or intensive searching could be mistaken for a graduate who benefited from AI skills. Accounting for such factors does not establish causation, but it makes the estimated relationship between AI proficiency and employment opportunities more credible.

Related Studies

Anthropic (2026): Labor Market Impacts of AI

Massenkoff and McCrory (2026), in Anthropic's study entitled Labor Market Impacts of AI: A New Measure and Early Evidence, introduced the concept of observed exposure. Their measure combines the theoretical ability of large language models to perform tasks with evidence of actual work-related AI use. The researchers give greater weight to tasks performed through automation than to tasks where AI is used only to assist a worker. They also consider how much of an occupation's overall work is represented by the affected tasks.

The study identifies market research analysts and marketing specialists among the most exposed occupations, with an observed exposure estimate of 64.8%, placing the occupation fifth in the reported top-ten list. This result is relevant to Marketing Management

because market research and marketing specialist responsibilities include preparing reports, presenting findings, and translating information into written material. Such activities overlap with the capabilities and observed work-related uses of generative AI. The result supports the view that marketing-related work is highly exposed to AI-assisted task change.

The researchers also emphasize that AI has not yet reached its full theoretical capability in the workplace. Their analysis found no systematic increase in unemployment among workers in highly exposed occupations since late 2022, although it found suggestive evidence that hiring among younger workers had slowed in exposed occupations. The study is relevant to the present research because it connects occupational task exposure with employment outcomes. It also supports the need to distinguish between the exposure of marketing tasks and the actual availability of jobs for Marketing Management graduates.

Mahoney, McEntarfer, and Wahal (2026): What Is Really Happening to Jobs?

Mahoney, McEntarfer, and Wahal (2026), in the Stanford Institute for Economic Policy Research policy brief *What Is Really Happening to Jobs? Separating AI Hype from Reality*, reviewed available evidence on AI and employment. The authors explain that the overall employment effects of AI appear limited so far, although the difficult labor market faced by recent graduates may be partly related to AI. They also report that AI's effect on worker productivity is mixed but generally positive, that business adoption has accelerated unevenly, and that current evidence is not enough to determine the full long-term effects of AI.

The brief notes that routine research, analysis, and writing tasks commonly found in junior roles can now be performed with AI assistance. It discusses evidence of weaker hiring for younger workers in some AI-exposed occupations, while also explaining that other factors may contribute to this pattern. The authors caution against treating AI as the sole explanation for every change in employment. This study is relevant to the present research because Marketing Management graduates often seek entry-level roles that include research, content preparation, and basic campaign analysis. It suggests that AI may change the expectations attached to these positions, even when broad job displacement is not evident.

The policy brief further explains that AI can improve the speed or productivity of some tasks, but results vary according to the task and the user's ability to evaluate the output. This supports the importance of AI literacy and human judgment. For graduates, the ability to use AI effectively may create an advantage, while the inability to verify or appropriately apply AI-generated information may limit the value of the tool.

Brynjolfsson, Chandar, and Chen (2026): Canaries in the Coal Mine?

Brynjolfsson, Chandar, and Chen (2026), in the revised Stanford Digital Economy Lab working paper *Canaries in the Coal Mine? Six Facts about the Recent Employment Effects of Artificial Intelligence*, analyzed high-frequency payroll data from ADP covering millions of U.S. workers through June 2026. The researchers reported that they found no evidence of widespread economy-wide job displacement. However, employment among young workers aged 22 to 25 in AI-exposed occupations was 19% below the level it would have reached if it had kept pace with less-exposed peers. The researchers found that this difference was mainly associated with reduced hiring rather than increased worker separations.

The study also found that employment declines were concentrated in occupations where AI use primarily substituted for human tasks. In occupations where AI mainly complemented workers, employment was flat or rising, particularly among experienced workers. The authors describe their findings as early descriptive indicators rather than definitive causal estimates. They also identify limitations, including differences between the ADP sample and national survey benchmarks and the fact that some employment trends began before generative AI became widely available.

This study is related to the present research because it highlights a possible challenge for new graduates: AI may influence access to entry-level opportunities before it produces visible large-scale unemployment. Although the study focuses on U.S. workers and does not specifically examine Filipino Marketing Management graduates, it provides a reason to examine hiring criteria, AI-related competencies, and the employment experiences of graduates entering marketing-related positions.

Green (2024): Artificial Intelligence and the Changing Demand for Skills in the Labour Market

Green (2024), in an OECD working paper, examined how AI changes the demand for skills in occupations that do not require specialized AI expertise. The study found that management and business skills were among the most demanded in occupations with high AI exposure. These include project management, finance, administration, and clerical-related skills. The report also observed an increase over time in vacancies requiring emotional, cognitive, or digital skills, while its establishment-level analysis indicated that demand for some of these skills may be beginning to decline in certain contexts.

The study is relevant because Marketing Management is a business-oriented field in which graduates may use AI without needing to become AI developers. Instead, they may need to combine business understanding with digital skills and the ability to assess AI-assisted work. The OECD findings support the inclusion of AI competencies, management abilities, and broader workplace skills when examining employer requirements for marketing-related positions.

Cucio and Hennig (2025): Artificial Intelligence and the Philippine Labor Market

Cucio and Hennig (2025), in the International Monetary Fund working paper *Artificial Intelligence and the Philippine Labor Market: Mapping Occupational Exposure and Complementarity*, examined AI exposure using Philippine labor force survey data. Their analysis found that around one-third of workers were in occupations highly exposed to AI. Of those highly exposed jobs, approximately 61% were also rated as highly complementary, suggesting that AI may support workers and increase productivity rather than replace them. The study also identified college-educated, young, urban, female, and higher-paid service-sector workers as groups with greater exposure.

The researchers identified business process outsourcing as a sector with a high proportion of jobs at risk of displacement, while emphasizing the importance of workforce reskilling, digital skills, infrastructure, and appropriate policies. The study is particularly relevant because it examines the Philippine labor market rather than relying exclusively on data from other countries. Although it does not isolate Marketing Management graduates as a

separate group, its findings show that AI exposure is a local labor-market issue and that exposure and complementarity must be considered together.

For the present study, the IMF findings provide a Philippine context for examining marketing employment. They suggest that AI may both create productivity opportunities and place pressure on certain tasks or positions. The study therefore supports investigating whether graduates' AI skills are associated with employment access and how graduates perceive the opportunities and threats created by AI.

Synthesis of the Reviewed Literature and Studies

The reviewed literature and studies show that artificial intelligence is changing work primarily through its effects on tasks, skills, and hiring practices. The findings do not support the idea that every occupation exposed to AI will automatically disappear. Instead, AI can automate selected activities, assist employees with existing responsibilities, and change the skills required to perform a job. This distinction is central to understanding its possible effects on Marketing Management graduates.

The Anthropic (2026) study provides direct evidence that market research analysts and marketing specialists are among the occupations with high observed AI exposure. The 64.8% estimate indicates substantial overlap between marketing-related tasks and work activities being performed with AI. This can be explained by the field's reliance on research, reporting, written communication, information analysis, and content preparation. These tasks can be supported or partly automated by generative AI. However, the estimate measures task exposure and should not be interpreted as a forecast that the same percentage of marketing jobs will be lost.

The Stanford SIEPR brief and the Stanford Digital Economy Lab study add an employment dimension to this issue. They indicate that broad economy-wide job displacement has not been established, but that entry-level hiring in some AI-exposed occupations may be weakening. This raises a concern for new graduates who commonly begin with research, writing, and basic analysis responsibilities. At the same time, both studies caution that other economic and organizational factors may contribute to observed hiring changes. Therefore, the relationship between AI and employment opportunities should

be examined rather than assumed.

The OECD study emphasizes that AI-related change affects the demand for skills even in jobs that do not require specialized AI development. Business, management, cognitive, emotional, and digital skills remain relevant to the changing workplace. The IMF study provides a Philippine perspective by showing that many locally exposed occupations are also complementary to AI. Together, these findings suggest that graduates may benefit from combining AI literacy with marketing knowledge, critical evaluation, communication, creativity, and sound decision-making.

Despite these contributions, the reviewed studies do not directly establish how AI skills affect the employment opportunities of Filipino Marketing Management graduates. The international studies largely use U.S. occupational or employment data, while the Philippine study examines the broader labor market rather than this particular graduate group. In addition, the reviewed studies work at the level of occupations rather than measuring individual graduates' AI proficiency, and they do not account for differences in candidate strength and job search effort. There is therefore a need for research that considers employer requirements in marketing-related job postings alongside the experiences and perceptions of Marketing Management graduates, measures their AI proficiency through performance rather than self-report, and examines employment outcomes while accounting for candidate strength and job search intensity. The present study addresses this need.

CHAPTER 3

METHODOLOGY

This chapter describes how the study will be conducted. It presents the research design, the research locale, the sources of data and how they will be sampled, the variables, the instruments, the procedures for establishing their validity and reliability, the data gathering procedure, the ethical considerations, and the statistical treatment of the data. Every research question and hypothesis stated in Chapter 1 is matched to a data source, a measure, and a statistical test, so that each thing the study sets out to measure has a defined method.

Research Design

The study will use a convergent mixed-methods design with two parallel strands whose findings are combined at the interpretation stage. The first strand is a quantitative content analysis of job postings, using a descriptive-comparative approach that contrasts marketing-related postings published before and after the public release of ChatGPT on November 30, 2022. This strand captures what employers ask for. The second strand is a cross-sectional survey of Marketing Management graduates that uses a correlational, predictive approach. It captures what graduates can demonstrably do with AI, how their job search has gone, and how they perceive their situation.

The two strands are used together because each covers a weakness of the other. Job postings show employer requirements objectively but say nothing about whether graduates who meet them are hired. The survey shows graduate outcomes and perceptions but reflects only the respondents' own experiences. Considering both allows the findings on employer demand, graduate outcomes, and graduate perceptions to be compared and interpreted together (see Figure 3). Table 1 shows how each research question is matched to its data source, measures, and statistical treatment.

Two features of the survey strand respond directly to concerns raised in Chapters 1 and 2. First, AI skills will be measured through a scored performance test, not through self-rating, to obtain a more objective estimate of what each graduate actually knows and can do. Second, because time to employment also depends on how strong a candidate a graduate is and how actively the graduate looks for work, these two factors will be measured and

statistically controlled. The design is observational. It can show whether AI proficiency is associated with employment outcomes after accounting for these factors, but it cannot by itself prove cause and effect.

Figure 3

Research Design and Flow of the Study

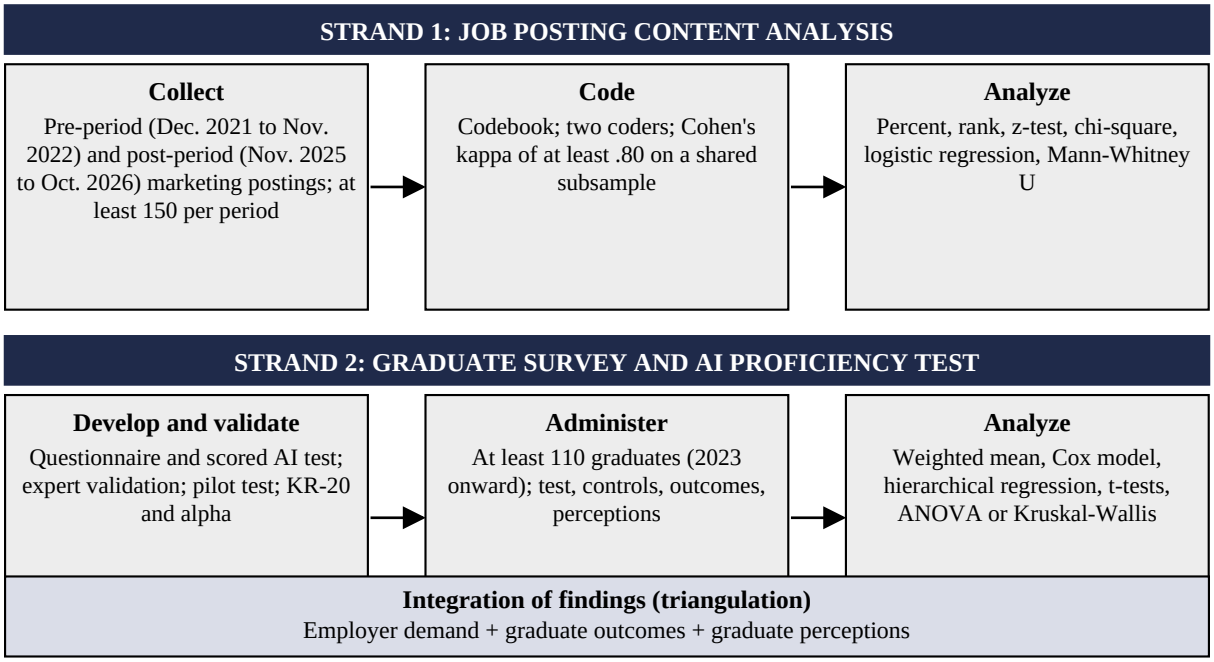


Table 1

Alignment of Research Questions, Data Sources, Measures, and Statistical Treatment

Research question	Data source	Main measures	Main statistical treatment
RQ1: Employer demand for AI skills ($H0_1$)	Job postings (pre vs. post)	Presence of AI skill requirement; AI competency categories; role family; industry	Percentage and rank; two-proportion z-test; chi-square test; binary logistic regression
RQ2: AI skills and employment opportunities ($H0_2$)	Graduate survey and AI proficiency test	AI Proficiency Score; time to employment; interview rate; offer rate; degree relevance; candidate strength; job search intensity; employers' use of AI skills in hiring	Weighted mean; Cox proportional hazards regression; hierarchical linear regression
RQ3: Effect of AI on employment opportunities ($H0_3$)	Graduate survey; job postings	Tasks automated, augmented, or newly created; AI-Driven Change Index; years of experience, education, and portfolio requirements in postings	Frequency and percentage; Cox and hierarchical regression; Mann-Whitney U; chi-square test
RQ4: Graduates' perceptions ($H0_4$)	Graduate survey	AI preparedness; perceived opportunities; perceived threats	Weighted mean and SD; one-sample and paired *t*-tests; Mann-Whitney U, Kruskal-Wallis, or ANOVA

Note. RQ = research question; H0 = null hypothesis.

Research Locale

The study is set in the Philippines. Job postings will be gathered from online job platforms that serve Filipino job seekers, including JobStreet, LinkedIn, Indeed Philippines, Kalibrr, and OnlineJobs.ph, together with the career pages of employers where dated copies are available. Postings that are open to Filipino applicants for remote work with overseas employers will also be included, consistent with the scope of the study. The final list of platforms will depend on the availability of dated and archived postings. Survey respondents will be reached online through the alumni networks of the Marketing Management program of Palawan State University, Brookes Point Campus, other Philippine universities offering the program, and professional and job-seeker groups on social media and LinkedIn.

Population and Sampling

Job Postings

The population consists of publicly available online postings for marketing-related positions in the Philippines and remote positions open to Filipinos. The unit of analysis is one

posting. The pre-period runs from December 1, 2021, to November 29, 2022, the twelve months before the release of ChatGPT. The post-period runs from November 1, 2025, to October 31, 2026, the most recent twelve months. Pre-period postings will be retrieved from dated platform archives, the Internet Archive's Wayback Machine, and saved copies. Only postings with a verifiable posting date within the period will be used.

A posting is eligible when it meets the criteria in Table 2. Postings will be selected through stratified purposive sampling across the marketing role families in the codebook, with at least 15 eligible postings per role family per period where available. Where more eligible postings exist than needed, postings will be chosen within each role family by simple random selection using a random number generator. The target is at least 150 eligible postings per period (at least 300 in total). With 150 postings per period, a one-tailed two-proportion test at the .05 level has about 80% power to detect an increase in the share of postings requiring AI skills from 10% to about 21%. If fewer than 150 eligible postings can be found for a period, all eligible postings will be used and the reduced power will be reported.

Table 2

Eligibility Criteria for Job Postings

Criterion	Included	Excluded
Position	Duties align with Marketing Management (market research, digital marketing, brand management, advertising, campaign planning, marketing analysis, content marketing)	Sales-only, customer service-only, general administrative, or graphic-design-only roles with no marketing function
Location	Based in the Philippines, or remote and open to Filipino applicants	Postings limited to non-Filipino applicants or to other countries
Date	Verifiable posting date within the pre-period or post-period	Undated postings or postings outside both periods
Content	States duties and qualifications	Postings with no description beyond a job title
Uniqueness	First instance of a posting	Duplicates, reposted or cross-listed copies of the same vacancy

Survey Respondents

The population is Filipino graduates of Marketing Management programs. The sampling frame is limited to graduates from 2023 onward, so that their job search took place after the release of ChatGPT. Respondents must be currently working in a marketing-related

position aligned with Marketing Management or actively seeking one. Graduates employed only in unrelated work, those not seeking marketing-related work, and graduates before 2023 are excluded. Because no complete list of such graduates exists, the study will use purposive sampling combined with snowball sampling, in which respondents are asked to refer other eligible graduates.

The target sample is at least 110 valid respondents, with an aim of 120 to allow for invalid responses. This number was set from the most demanding analysis. A hierarchical regression with up to eight predictors, a medium effect size ($f^2 = .15$), a .05 level of significance, and 80% power requires about 109 respondents (Faul et al., 2009). A correlation of $r = .30$ at the same power requires about 85. The Cox model uses four predictors, which calls for about 40 employment events under the common guideline of ten events per predictor. With at least 110 respondents, and as long as roughly four in ten of them have already found a marketing-related job, this guideline is met. Responses will be set aside if the test is incomplete, if both attention-check items are failed, or if the same person responds twice.

Variables of the Study

Table 3 lists every variable the study measures, its role in the analysis, how it is measured, and its level of measurement. Candidate strength, job search intensity, and graduation year are control variables. Their purpose is to prevent a graduate who found work quickly because of strong qualifications or intensive searching from being mistaken for a graduate who benefited from AI skills.

Table 3

Variables, Their Roles, Measurement, and Level of Measurement

Variable	Role	Measurement	Level
Time period	Independent (RQ1)	Pre-period or post-period posting	Nominal
Marketing role family	Independent (RQ1)	Six role families in the codebook	Nominal
Industry	Independent (RQ1)	Industry of the hiring organization, from codebook categories	Nominal
AI skill requirement	Dependent (RQ1)	Present (1) or absent (0) in a posting; AI competency categories coded	Nominal
AI Proficiency Score	Independent (RQ2)	Percentage of points earned on the scored AI proficiency test (0 to 100)	Ratio

Variable	Role	Measurement	Level
Candidate Strength Index	Control (RQ2, RQ3)	Mean of the standardized scores of GWA, OJT or internship, certifications, portfolio, and prior relevant experience	Interval (index)
Job Search Intensity Index	Control (RQ2, RQ3)	Mean of the standardized scores of applications per month, search hours per week, and number of search channels	Interval (index)
Graduation year	Control (RQ2, RQ3)	Year of graduation (2023 to 2026)	Ordinal
Time to employment	Dependent, primary (RQ2, RQ3)	Weeks from graduation or start of search to first marketing-related job; censored if still seeking	Ratio (censored)
Interview rate and offer rate	Dependent, secondary (RQ2, RQ3)	Interviews (or offers) divided by applications submitted, times 100	Ratio
Degree relevance	Dependent, secondary (RQ2, RQ3)	Mean of a four-item, five-point scale on how closely the first position uses the degree	Interval (scale)
Employers' use of AI skills in hiring	Descriptive (RQ2a)	Mean of a six-item, five-point scale	Interval (scale)
Task-level AI effect	Descriptive (RQ3a)	For each of 12 marketing tasks: not affected, augmented, largely automated, or new task created	Nominal
Posting requirements	Dependent (RQ3b)	Years of experience, education, and portfolio requirements and AI-related duties in postings	Ratio and nominal
AI-Driven Change Index	Independent (RQ3)	Mean of an eight-item, five-point scale on AI-related change in tasks and hiring criteria	Interval (scale)
AI preparedness	Descriptive and grouping (RQ4)	Mean of an eight-item, five-point self-report scale	Interval (scale)
Perceived opportunities; perceived threats	Dependent (RQ4)	Means of two six-item, five-point subscales	Interval (scale)

Research Instruments

Three instruments will be used: a codebook for the job postings, a graduate questionnaire, and a scored AI proficiency test that forms part of the questionnaire. The questionnaire and test are reproduced in Appendix A, the answer key and scoring rubric in Appendix B, the codebook in Appendix C, and the expert validation form in Appendix D.

Job Posting Codebook

Each posting will be recorded on a coding sheet. The sheet captures the posting date, platform, role family, industry, location type, years of experience required, education requirement, whether a portfolio is required, and whether the posting is entry-level (one year of experience or less, or stated as open to fresh graduates). It also records whether the posting contains an AI-related competency, whether that competency is stated as required or only preferred, which of the six categories in Table 4 apply, any named AI tools, and any AI-related duties that represent new tasks. A posting is coded as requiring AI skills when it lists at least one competency in Table 4 as required or preferred. A second, stricter indicator counts only competencies stated as required. Coders will read each posting and apply the category definitions, with a keyword list used only to locate candidate mentions, so that terms used in a non-AI sense are not counted.

Table 4

AI Skill Categories in the Job Posting Codebook

Category	Definition	Example wording
Tool proficiency	Use of named generative AI tools	"ChatGPT," "Gemini," "Copilot," "Midjourney"
Prompting	Writing and refining instructions for AI systems	"prompt writing," "prompt engineering"
AI-assisted content creation	Producing or editing copy, images, or video with AI	"AI-generated content," "AI copywriting tools"
AI-assisted research and analytics	Using AI for market research, summarizing, or data interpretation	"AI-powered analytics," "AI research tools"
Evaluation and responsible use	Checking AI outputs; ethics, privacy, and quality control	"verify AI outputs," "responsible AI use"
General AI familiarity	Broad or unspecified AI or automation skills	"AI tools," "familiar with AI," "generative AI"

Graduate Questionnaire

The questionnaire has eight parts, summarized in Table 5, and will be answered online. Parts I to IV (profile, candidate strength and search, outcomes, and preparedness) come before the AI proficiency test in Part V, so that respondents' self-rated preparedness is not influenced by their experience of the test. Likert items use a five-point agreement scale (5 = strongly agree to 1 = strongly disagree). Two attention-check items are placed within the Likert parts.

Table 5

Structure of the Graduate Questionnaire

Part	Content	Items	Response format	Used for
I. Respondent profile	Graduation year, employment status, position, industry, location (Philippines or overseas)	8	Selection and short entry	Description; eligibility
II. Candidate strength and job search	GWA; OJT or internship months; certifications; portfolio; prior experience; applications per month; search hours per week; search channels	8	Numeric entry and selection	Control variables
III. Employment outcomes	Graduation and job start dates; total applications, interviews, offers; degree relevance scale	7 + 4	Dates, counts, and five-point scale	RQ2, RQ3 outcomes
IV. AI preparedness	Self-rated readiness to use AI in marketing work	8	Five-point scale	RQ4a
V. AI proficiency test	Knowledge, prompt evaluation, output verification, applied tasks	40 points	Scored; see Table 6	RQ2 predictor
VI. Employers' use of AI skills in hiring	Whether AI skills were asked for, tested, or valued in applications and interviews	6	Five-point scale	RQ2a
VII. AI-driven change	A: effect of AI on 12 marketing tasks. B: change in hiring criteria and entry-level expectations	12 + 8	A: four categories. B: five-point scale	RQ3a, RQ3b; AI-Driven Change Index
VIII. Perceived opportunities and threats	Opportunities subscale and threats subscale	6 + 6	Five-point scale	RQ4b

AI Proficiency Test

AI proficiency is measured as ability, not as confidence. Self-ratings are known to be unreliable (Kruger & Dunning, 1999), so respondents will answer questions that have correct or incorrect answers, or that are scored against a rubric, as in a short examination. The test contains four parts, shown in Table 6, for a total of 40 points. Items are set in marketing situations and test lasting concepts (what AI can and cannot do, how to instruct it, how to check it, and how to use it responsibly) so that results do not depend on any one product or feature. The percentage of the 40 points earned is the AI Proficiency Score.

Table 6

Composition and Scoring of the AI Proficiency Test

Part	What it tests	Items	Points	Scoring
A. Knowledge and limits	Capabilities, limitations, hallucination, data privacy, appropriate use	10 multiple-choice	10	1 point per correct answer
B. Prompt evaluation	Recognizing the stronger of two prompts for a marketing task	6 items	6	1 point per correct choice
C. Output verification	Finding planted errors in AI-written marketing text	4 passages, 3 planted errors each	12	1 point per error identified
D. Applied tasks	Writing a prompt and a checking plan for a realistic marketing task	2 tasks	12	Rubric score of 0 to 6 per task, by two raters
Total			40	AI Proficiency Score = points earned / 40 x 100

The test will be taken closed-book and without AI tools or web search, within a 30-minute limit, with the order of items randomized within each part. Respondents confirm before starting that they will not use outside help. The two applied tasks in Part D are scored independently by two raters who do not see respondents' identities, using a six-point rubric that awards one point each for a clear goal, stated audience and context, specified constraints and format, a plan to verify the output, attention to privacy or ethics, and sound marketing judgment.

Scoring and Interpretation

Responses to Likert items will be interpreted with the scale in Table 7. Subscale scores are the mean of their items. The Candidate Strength Index and the Job Search Intensity Index are formed by converting each component to a standardized (*z*) score, reversing GWA so that higher always means stronger, and averaging the components. Standardizing puts measures with different units on a common scale, and combining them keeps the number of predictors small relative to the sample.

Table 7

Interpretation of Mean Scores on the Five-Point Scale

Mean range	Verbal interpretation
4.21 to 5.00	Strongly agree (very high)
3.41 to 4.20	Agree (high)
2.61 to 3.40	Neutral (moderate)
1.81 to 2.60	Disagree (low)
1.00 to 1.80	Strongly disagree (very low)

Validity and Reliability of the Instruments

Content validity will be established by a panel of five experts: two Marketing Management faculty members, one practitioner experienced in AI use in marketing, one research or statistics expert, and one language or measurement specialist. Each expert will rate every item for relevance on a four-point scale using the form in Appendix D. The item-level content validity index (I-CVI) is the proportion of experts rating an item 3 or 4. With five experts, every retained item should reach an I-CVI of 1.00, and the scale-level average should be at least .90 (Lynn, 1986). Items falling short will be revised or removed, and the panel's comments on wording, clarity, and difficulty will be applied.

A pilot test will be conducted with 20 to 30 Marketing Management graduates who will not be part of the main sample. The pilot will check clarity, completion time, and the functioning of the online form. For the AI proficiency test, item difficulty (the proportion answering correctly) and discrimination (the difference in correct answers between the top and bottom 27% of scorers) will be examined, and items with difficulty outside .20 to .90 or discrimination below .20 will be revised or removed.

Reliability of the right-or-wrong items in Parts A to C of the test will be assessed with the Kuder-Richardson Formula 20 (Kuder & Richardson, 1937), and reliability of the Likert scales with Cronbach's alpha (Cronbach, 1951). A value of .70 or higher will be treated as acceptable (Nunnally & Bernstein, 1994). Agreement between the two raters on Part D will be assessed with the two-way random-effects intraclass correlation coefficient for single raters (Shrout & Fleiss, 1979), with a target of at least .75. Because the Candidate Strength and Job Search Intensity indices combine different measures into an index and do not form a

unidimensional scale, internal consistency is not applied to them.

For the job postings, two coders will independently code a random 20% subsample of at least 60 postings. Agreement on the main coding decisions will be measured with Cohen's kappa (Cohen, 1960), and a value of at least .80 will be required (Landis & Koch, 1977). If agreement is lower, the codebook will be clarified and the subsample recoded. Remaining disagreements will be settled by discussion, and the rest of the postings will then be divided between the coders.

Data Gathering Procedure

Data gathering will proceed in four phases:

1. **Preparation.** The codebook, questionnaire, and AI proficiency test will be drafted, reviewed by the expert panel, revised, pilot-tested, and finalized. Permission to share the survey will be requested from program heads, alumni officers, and group administrators.
2. **Job posting collection.** Eligible postings for the pre-period and post-period will be located, screened against Table 2, sampled, and saved with their dates and sources. Coders will then code the postings using the codebook.
3. **Survey administration.** The online survey will be distributed through alumni networks, social media groups, and referrals. Respondents will read the consent statement, confirm eligibility, complete Parts I to IV, take the timed test in Part V, and complete Parts VI to VIII. Reminders will be sent during the collection period.
4. **Data preparation.** Responses will be screened for completeness, duplicates, and attention-check failures. The test will be scored, Part D will be rated by two raters, and indices and composite scores will be computed.

Ethical Considerations

The study will follow the Data Privacy Act of 2012 (Republic Act No. 10173) and the research ethics procedures of the university. Participation is voluntary. The consent statement at the start of the questionnaire explains the purpose of the study, what respondents will be asked to do, the approximate time required, the right to skip questions or stop at any time, and how responses will be used. Only information needed for the study will be collected.

Respondents will not be asked for their names, and any contact details needed for follow-up will be stored apart from their answers. Individual responses and test scores will not be disclosed, and results will be reported only in aggregate. Data will be kept in a password-protected file accessible only to the researchers and deleted after the study period required by the institution. Job postings are publicly available, no personal data will be collected from them, and employers will not be named in the report. The AI proficiency test is used only for research and will not be used to evaluate respondents' employability.

Statistical Treatment of Data

All inferential tests will use a .05 level of significance. Analyses will be run in JASP or SPSS, and the Cox model and survival curves may be produced in R. The main tests for each hypothesis and the decision rules are shown in Table 8.

Research Question 1

The frequency and percentage of postings containing each AI competency category will be computed for each period and ranked to identify the most frequently required competencies (RQ1a). To test $H0_1$, the share of postings requiring AI skills in the post-period will be compared with the share in the pre-period using a one-tailed two-proportion *z*-test, with Cohen's *h* as the effect size (Cohen, 1988) and a 95% confidence interval for the difference. Percent change in the number of eligible postings found per period will be reported descriptively, with the caution that it depends on how complete each platform's archive is. For RQ1b, a chi-square test of independence (or Fisher's exact test when expected counts are below 5) will examine whether AI skill requirements differ by role family and by industry, with Cramér's *V* as effect size. A binary logistic regression with period, role family, and industry as predictors will estimate their combined influence as odds ratios.

Research Question 2

The extent to which AI skills influence hiring decisions (RQ2a) will be described with the mean, standard deviation, and verbal interpretation of Part VI, and compared with the frequency of AI requirements in the postings. For $H0_2$, the primary outcome is time to employment. Because graduates who are still searching have not yet reached the event, their times are right-censored. A Cox proportional hazards model handles this correctly by using

what is known about those graduates up to the survey date. Candidate Strength Index, Job Search Intensity Index, and graduation year will be entered first, and the AI Proficiency Score added second. The addition is tested with the likelihood-ratio test, and the effect is reported as a hazard ratio with a 95% confidence interval. A hazard ratio above 1 means faster employment per unit increase in the predictor. Kaplan-Meier curves (Kaplan & Meier, 1958) will compare time to employment for respondents with low, middle, and high AI Proficiency Scores (lowest third, middle third, highest third). The proportional hazards assumption will be checked with Schoenfeld residuals.

To identify which employment opportunities are associated with AI skills (RQ2b), the secondary outcomes (interview rate, offer rate, and degree relevance) will each be analyzed with hierarchical linear regression. The controls will enter in Step 1 and the AI Proficiency Score in Step 2, and the change in R^2 will be tested. Because three secondary outcomes are tested, *p*-values will be adjusted with the Holm procedure (Holm, 1979). Interview and offer rates are used instead of raw counts, since graduates who apply more often receive more interviews regardless of their AI skills.

Research Question 3

For RQ3a, the number and percentage of respondents classifying each of the 12 marketing tasks as not affected, augmented, largely automated, or a new task will be reported, and tasks will be ranked by the percentage in each category. Tasks appearing in postings as AI-related duties after the release of ChatGPT will be listed as evidence of newly created tasks. For RQ3b, years of experience required in postings will be compared before and after using the Mann-Whitney *U* test, and the share of entry-level postings and the share requiring a degree or a portfolio will be compared using two-proportion *z*-tests or chi-square tests. Part VII-B responses will be summarized with weighted means.

For $H0_3$, the same Cox and hierarchical regression models used for RQ2 will be fitted with the AI-Driven Change Index as the predictor, after the same controls. A model including both the AI Proficiency Score and the AI-Driven Change Index will also be reported, and variance inflation factors will be checked to be sure the two predictors do not overlap too heavily (VIF below 5). The before-and-after tests on the postings serve as corroborating

evidence on whether hiring criteria changed.

Research Question 4

Preparedness, perceived opportunities, and perceived threats will be summarized with weighted means, standard deviations, and the verbal interpretation in Table 7. To test whether graduates perceive significant opportunities and threats ($H0_4$), each subscale mean will be compared against the neutral midpoint of 3.00 with a one-sample *t*-test (one-tailed, testing for a mean above 3.00), and a paired-samples *t*-test will compare the opportunities and threats subscales with each other. To test whether preparedness varies, scores will be compared across employment status (employed in the field or still seeking) using an independent-samples *t*-test, across graduation year and frequency of AI use using one-way ANOVA, and the nonparametric alternatives will be used when assumptions are not met. Self-rated preparedness will also be correlated with the AI Proficiency Score to show how closely graduates' confidence matches their demonstrated ability.

Table 8

Tests of the Hypotheses and Decision Rules

Hypothesis	Statistical test	Decision rule
$H0_1$	One-tailed two-proportion *z*-test on the share of postings requiring AI skills, post-period vs. pre-period	Reject $H0_1$ if $p < .05$ and the post-period share is higher
$H0_2$	Cox regression of time to employment: Step 1 controls, Step 2 AI Proficiency Score; likelihood-ratio test. Secondary: hierarchical regression with Holm adjustment	Reject $H0_2$ if the AI Proficiency Score contributes significantly at $p < .05$ in the primary model
$H0_3$	Same models with the AI-Driven Change Index as predictor; supporting posting tests (Mann-Whitney *U*, chi-square)	Reject $H0_3$ if the AI-Driven Change Index contributes significantly at $p < .05$
$H0_4$	One-sample *t*-tests of opportunities and threats against 3.00; *t*-test, ANOVA, or nonparametric tests of preparedness across groups	Reject $H0_4$ if both subscale means are significantly above 3.00 and preparedness differs significantly across at least one grouping

Assumptions and Data Screening

Before inferential tests, data will be checked for missing values, outliers, and duplicates. Normality will be assessed with the Shapiro-Wilk test and inspection of plots, equality of variances with Levene's test, and multicollinearity with variance inflation factors.

When assumptions are not met, the nonparametric counterparts will be used (Wilcoxon signed-rank test, Mann-Whitney *U*, Kruskal-Wallis, Spearman's rho). If less than 5% of values are missing on a variable, cases will be dropped for that analysis, and if more, multiple imputation will be used for predictors and sensitivity analyses will be reported. Time-to-employment will not be imputed. As a sensitivity check, the models will be repeated with each component of the Candidate Strength and Job Search Intensity indices entered separately.

Formulas and Worked Examples

Table 9 shows the main computations. Each formula uses descriptive names for its parts, and each is followed by a worked example with illustrative numbers, not results from the study.

Table 9

Main Formulas, Their Parts, and Worked Examples

Measure	Formula and meaning of each part	Worked example (illustrative)
Percentage	percentage = (number_of_postings_with_AI_skill / total_postings_in_period) x 100. *number_of_postings_with_AI_skill* counts postings coded as requiring AI skills; *total_postings_in_period* is every eligible posting in that period.	54 of 150 postings: (54 / 150) x 100 = 36.0%
Weighted mean	weighted_mean = sum of (number_of_responses_at_rating x rating) / total_respondents. *rating* is the scale value (1 to 5); *number_of_responses_at_rating* is how many respondents chose it.	Of 100 respondents: 20 chose 5, 40 chose 4, 25 chose 3, 10 chose 2, 5 chose 1. (100 + 160 + 75 + 20 + 5) / 100 = 3.60, interpreted as Agree
AI Proficiency Score	ai_proficiency_score = (points_earned / total_possible_points) x 100. *points_earned* is the sum of Parts A to D; *total_possible_points* is 40.	31 points earned: (31 / 40) x 100 = 77.5
Interview rate (offer rate is the same with offers)	interview_rate = (interviews_received / applications_submitted) x 100. *interviews_received* is the number of interview invitations; *applications_submitted* is the number of applications sent.	6 interviews from 40 applications: (6 / 40) x 100 = 15.0%
Standardized score (used in the indices)	standardized_score = (person_value - group_average) / group_standard_deviation. *person_value* is one respondent's value on a component; the average and standard deviation come from all respondents. The index is the mean of a respondent's standardized scores.	Five components with standardized scores of +0.8, +0.2, -0.1, +0.6, 0.0 give an index of (0.8 + 0.2 - 0.1 + 0.6 + 0.0) / 5 = 0.30

Measure	Formula and meaning of each part	Worked example (illustrative)
Two-proportion z-test ($H0_1$)	z = (share_after - share_before) / standard_error, where standard_error = square root of [pooled_share x (1 - pooled_share) x (1/postings_after + 1/postings_before)] and pooled_share = (AI_postings_before + AI_postings_after) / (postings_before + postings_after).	12 of 150 before (.08), 54 of 150 after (.36): pooled_share = 66 / 300 = .22; standard_error = square root of [.22 x .78 x (1/150 + 1/150)] = .0478; z = (.36 - .08) / .0478 = 5.85
Cox model ($H0_2$, $H0_3$)	hazard_of_employment_at_week = baseline_hazard_at_week x exp(effect_of_AI_score x ai_proficiency_score + effect_of_candidate_strength x candidate_strength_index + effect_of_search_intensity x search_intensity_index + effect_of_graduation_year x graduation_year), where exp(value) means e raised to that value. *hazard* is the chance of getting the first marketing-related job in a given week among those still without one; each *effect_of* term is the weight the model estimates; exp of an effect is the hazard ratio.	If the model estimates effect_of_AI_score = 0.018 per point, a 10-point rise gives exp(0.018 x 10) = exp(0.18) = 1.20: a 10-point higher score goes with a 20% higher weekly chance of being hired, holding the controls constant

Methodological Limitations

Several limitations follow from the design and are noted so that the findings are read appropriately. Respondents are recruited by purposive and snowball sampling, so the sample will not statistically represent all Marketing Management graduates. Dates and counts of applications, interviews, and offers are self-reported and rely on memory. The AI proficiency test will be taken online without direct supervision, so some respondents might use outside help despite the instructions, and the timed format and the declaration only reduce this risk. Pre-period postings depend on archives that may be incomplete, and the number of postings is affected by each platform's coverage. Finally, the study is observational, so any association between AI proficiency and employment outcomes, even after controlling for candidate strength and job search intensity, should not be interpreted as proof that AI skills caused the outcomes.

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Appendix A

Graduate Questionnaire and AI Proficiency Test

Informed Consent

You are invited to take part in a study on how artificial intelligence relates to the employment opportunities of Marketing Management graduates, conducted by BSBA Marketing Management researchers of Palawan State University, Brookes Point Campus. You will answer questions about your background, job search, and views, and take a timed test of AI knowledge and skills. It takes about 45 minutes. Taking part is voluntary. You may skip any question or stop at any time. We will not ask for your name, your answers will be reported only in combined form, and your data will be handled under the Data Privacy Act of 2012 (Republic Act No. 10173). The test is for research only and is not used to judge your employability. By selecting "I agree," you confirm that you are 18 or older, you have read this statement, and you agree to take part. For Part V, you also agree not to use AI tools, search engines, or other people's help.

[ ] I agree and wish to continue [ ] I do not agree

Part I. Respondent Profile

I-1. Year of graduation.

a. 2023
b. 2024
c. 2025
d. 2026

I-2. Degree program.

a. BSBA Major in Marketing Management
b. Another program with a Marketing Management major (please specify)

I-3. Current status.

a. Employed in a marketing-related position aligned with Marketing Management
b. Actively seeking a marketing-related position
c. Neither (survey ends; you are not eligible for this study)

I-4. Current or most recent marketing-related job title (if any).

I-5. Industry of your current or most recent employer, or the industry you are applying to.

a. Retail or consumer goods
b. IT-BPM or technology
c. Hospitality and tourism
d. Real estate or construction
e. Banking, finance, or insurance
f. Education or health
g. Media or advertising agency
h. Other (please specify)

I-6. Work location.

a. Philippines (on-site or hybrid)

b. Philippines (remote)
c. Overseas employer (remote)
d. Overseas (on-site)
e. Not yet employed

I-7. How often do you use generative AI tools for study or work?

a. Never
b. Rarely (less than monthly)
c. Monthly
d. Weekly
e. Daily

I-8. Province or region of residence.

Part II. Candidate Strength and Job Search

Answer for the period before you started your first marketing-related job. If you have not yet been hired, answer for your job search so far.

II-1. General weighted average (GWA) at graduation (1.00 highest to 5.00 lowest).

II-2. Total months of on-the-job training (OJT) or internship in a marketing-related setting.

II-3. Number of relevant certifications completed (for example, digital marketing or analytics).

II-4. Did you have a portfolio or work samples to show employers?

a. Yes
b. No

II-5. Months of prior relevant work experience (including freelance or part-time marketing work).

II-6. On average, how many job applications did you submit per month?

II-7. On average, how many hours per week did you spend searching and applying for jobs?

II-8. Which search channels did you use? (Select all that apply.)

a. Online job platforms
b. Company websites or career pages
c. Referrals or personal networking
d. Social media or professional networking sites
e. Recruitment or staffing agencies
f. Freelance or remote-work platforms
g. Walk-in or other (specify)

Part III. Employment Outcomes

III-1. Month and year of graduation.

III-2. Month and year you began actively searching for a marketing-related job (if before graduation, write the month and year you started).

III-3. Month and year you started your first marketing-related job (write "not yet" if you have not been hired).

III-4. Total number of applications submitted until your first marketing-related job (or until now, if still searching).

III-5. Total number of interview invitations received in that period.

III-6. Total number of marketing-related job offers received in that period.

III-7. Location of your first marketing-related job.

a. Philippines
b. Overseas (including remote for an overseas employer)
c. Not yet employed

Degree relevance. Rate each statement: 5 = Strongly agree, 4 = Agree, 3 = Neutral, 2 = Disagree, 1 = Strongly disagree. (If not yet employed, answer for the positions you are applying for.)

III-8. My first marketing-related position uses what I learned in my Marketing Management degree.

III-9. The main duties of the position match marketing functions such as research, advertising, branding, or campaign planning.

III-10. My degree was required or preferred for the position.

III-11. I consider the position to be aligned with my degree.

Part IV. AI Preparedness (Self-Report)

Rate each statement: 5 = Strongly agree, 4 = Agree, 3 = Neutral, 2 = Disagree, 1 = Strongly disagree.

IV-1. I understand what generative AI tools can and cannot do.

IV-2. I can choose the right AI tool for a given marketing task.

IV-3. I can write clear instructions that give useful results from AI tools.

IV-4. I can tell when an AI output is inaccurate or unsupported.

IV-5. I know how to use AI without exposing confidential information.

IV-6. I can use AI to speed up marketing tasks while keeping quality.

IV-7. I can adapt as AI changes how marketing work is done.

IV-8. I feel ready to meet employers' expectations about AI skills.

Part V. AI Proficiency Test

Instructions. This is a timed test (30 minutes). Do not use AI tools, search engines, notes, or help from others. Choose the best answer for Parts A and B. Item order is randomized by the online form. In Part A and B, the position of the correct option varies.

Part A. Knowledge and Limits (10 points; choose one answer for each)

A-1. Which statement best describes how a large language model produces text?

a. It looks up the exact answer in a database of verified facts.
b. It predicts likely next words based on patterns learned from large amounts of text.
c. It searches the internet in real time for every question.
d. It follows a fixed script written by programmers for each topic.

A-2. An AI tool gives you a confident statistic about a competitor's market share and cites a report you cannot find anywhere. This is best described as:

a. A database update
b. A copyright notice
c. A hallucination: fluent output that is not supported by real sources
d. A deliberate marketing strategy of the tool

A-3. Which of these is safest to enter into a public AI chatbot?

a. A customer list with names, emails, and purchase history
b. A client's unreleased pricing and launch plan
c. A general description of a fictional brand for practice
d. A signed supplier contract

A-4. You enter the same prompt twice and receive two different answers. The main reason is that:
a. Text generation involves some randomness in word selection.
b. The tool is malfunctioning.
c. The tool remembers that you disliked the first answer.
d. Prompts expire after a single use.

A-5. Which task is the least suitable to send out without human review?
a. Brainstorming 20 possible campaign slogans
b. Summarizing a long article for a team brief
c. Publishing a health claim about a product in an advertisement
d. Rewriting a caption in a more casual tone

A-6. An AI-written slogan is nearly identical to a competitor's slogan. The most important risk is:
a. Slower website loading
b. Lower email deliverability
c. Higher tax liability
d. Originality, trademark, and copyright problems

A-7. Why can AI-generated customer personas be misleading?
a. They are always built from a fresh survey of your target market.
b. They may reflect stereotypes and patterns from training data rather than your actual customers.
c. They cannot contain numbers.
d. They are legally invalid.

A-8. An AI answer is too general. What is the best first step to improve it?
a. Repeat the same prompt in capital letters.
b. Add details such as audience, goal, format, and constraints.
c. Tell it to "be correct."
d. Make the prompt shorter.

A-9. You ask an AI tool without live search access about this month's market trends. The main risk is that it:
a. Gives outdated or invented information as if it were current
b. Refuses to answer any question
c. Answers in another language
d. Charges a higher fee

A-10. When is it most appropriate to disclose that AI helped produce a marketing deliverable?
a. Never, because AI use is always private
b. Only when the output contains errors
c. Only when the tool is free
d. When company policy, a client agreement, the law, or audience trust calls for it

Part B. Prompt Evaluation (6 points; for each item, choose the stronger prompt, Prompt 1 or Prompt 2)

B-1. Task: social media post for a new iced coffee.

Prompt 1: Write a social media post for our product.

Prompt 2: Write a 40-word Facebook post for college students promoting our new iced coffee. Friendly tone, one call to action to visit our branch this weekend.

B-2. Task: summarizing customer feedback.

Prompt 1: Summarize the feedback in five bullet points grouped by theme (price, taste, service). Quote one short comment per theme and write "not mentioned" if a theme is missing.

Prompt 2: Summarize the feedback.

B-3. Task: estimating market share.

Prompt 1: Give me the market share numbers for the top three beverage brands in the Philippines.

Prompt 2: List the data I would need to estimate the market share of the top three beverage brands in the Philippines and where to find it. Do not invent any figures.

B-4. Task: email subject lines.

Prompt 1: Draft three subject lines under 50 characters announcing a 20% discount that ends Friday. No all-caps and no misleading urgency.

Prompt 2: Make a catchy email subject ASAP!!!

B-5. Task: revising an advertisement.

Prompt 1: Rewrite this ad to sound better.

Prompt 2: Rewrite this ad for parents aged 30 to 45. Keep the original claim exactly as written, shorten to 25 words, and use a friendly tone.

B-6. Task: reading a sales sheet.

Prompt 1: Tell me which product sold best and don't bother showing how you got the answer.

Prompt 2: Using the attached sales sheet, tell me which product sold best each quarter and show the numbers you used so I can check them.

Part C. Output Verification (12 points; each passage below was written by an AI tool and contains exactly three errors. List each error you find. One point per correct error.)

Passage C-1 (product description). Kape Brew Cold Brew is made from 100% Philippine arabica beans and contains no added sugar. Each 250 ml bottle has 25 grams of added sugar, which gives it a smooth, naturally sweet taste. Clinically proven to cure fatigue, it is the number one coffee brand in the world. Kape Brew is now available in 12 branches.

Passage C-2 (market research summary). In our survey of 200 respondents, 62% (124) preferred online shopping, 30% (60) preferred in-store shopping, and 18% (36) were undecided. The results show that most respondents favor in-store shopping. Because all respondents were university students, the findings can be applied to all Filipino consumers.

Passage C-3 (campaign recap). The four-week Facebook campaign ran from March 1 to March 14 with a budget of PHP 20,000. It reached 50,000 people and produced 1,000

link clicks, a click-through rate of 20% of the people reached. The cost per click was PHP 40.

Passage C-4 (pricing recommendation). Penetration pricing means launching a product at a high price to earn large margins from early adopters. We recommend this approach for our new budget snack because shoppers in this market are very price-sensitive. Since Brand X and Brand Y already compete in this market, it is a monopoly with a single dominant seller, so we should ignore competitor pricing.

Part D. Applied Tasks (12 points; type your answers)

D-1. Your manager asks you to use an AI tool to draft a one-week social media plan for a milk tea shop opening next month. (a) Write the exact prompt you would give the AI tool. (b) List how you would check and improve its output before using it.

D-2. You have 80 customer comments about a small hotel's breakfast service (price, food quality, speed, staff friendliness) and want AI to help summarize them for the owner. (a) Write your prompt. (b) State what information you would and would not paste into the tool, and how you would verify the summary.

Part VI. Employers' Use of AI Skills in Hiring

Rate each statement: 5 = Strongly agree, 4 = Agree, 3 = Neutral, 2 = Disagree, 1 = Strongly disagree. Select "not applicable" if the statement does not apply to your experience.

VI-1. Job postings I applied to listed AI skills as requirements or preferences.

VI-2. Interviewers asked about my experience in using AI tools.

VI-3. Employers asked me to demonstrate AI skills through a task or test.

VI-4. AI skills made my application more attractive to employers.

VI-5. Employers expected AI skills even when the posting did not state them.

VI-6. Employers asked how I check the accuracy of AI outputs.

VI-7. (Attention check) For this item, please select "Agree."

Part VII. AI-Driven Change in Marketing Tasks and Hiring Criteria

A. Tasks. For each marketing task, choose what best describes the effect of AI in the workplaces you have worked in or applied to: *Not affected*; *Augmented* (AI helps the worker); *Largely automated* (AI does most of the task); or *New task created* (AI has created a new related duty, for example building prompt workflows).

VII-A1. Writing advertising copy and promotional text

VII-A2. Writing social media captions and posts

VII-A3. Summarizing market research

VII-A4. Analyzing customer feedback and reviews

VII-A5. Reporting campaign performance

VII-A6. Keyword and SEO research

VII-A7. Creating images or graphics

VII-A8. Drafting emails and newsletters

VII-A9. Replying to customer inquiries

VII-A10. Analyzing and interpreting data

VII-A11. Planning campaigns and content

VII-A12. Making brand strategy and positioning decisions

If AI has created any new marketing tasks in the workplaces you know, please describe them:

_____

B. Hiring criteria and entry-level expectations. Rate each statement: 5 = Strongly agree, 4 = Agree, 3 = Neutral, 2 = Disagree, 1 = Strongly disagree.

VII-B1. Entry-level marketing positions now expect skills that used to be expected of more experienced staff.

VII-B2. Postings for entry-level marketing roles ask for AI skills.

VII-B3. Employers need fewer entry-level staff for routine writing and research tasks because of AI.

VII-B4. Employers expect new graduates to produce more work in less time using AI.

VII-B5. Portfolios or work samples have become more important in hiring.

VII-B6. Employers now give more weight to checking and judgment skills.

VII-B7. AI has changed the kinds of marketing tasks given to new hires.

VII-B8. Because of AI, the criteria used to select marketing applicants have changed.

Part VIII. Perceived Opportunities and Threats of AI

Rate each statement: 5 = Strongly agree, 4 = Agree, 3 = Neutral, 2 = Disagree, 1 = Strongly disagree.

Opportunities.

VIII-O1. AI will create new marketing jobs or roles.

VIII-O2. AI makes it easier to produce better marketing work.

VIII-O3. AI skills help me stand out to employers.

VIII-O4. AI opens opportunities with international employers.

VIII-O5. AI lets me take on higher-level work earlier in my career.

VIII-O6. AI lowers the cost of starting freelance or business work in marketing.

Threats.

VIII-T1. AI will reduce the number of entry-level marketing jobs.

VIII-T2. AI makes it harder for new graduates to compete for jobs.

VIII-T3. Employers may prefer using AI over hiring new graduates.

VIII-T4. My marketing degree may lose value because of AI.

VIII-T5. AI may lower pay for marketing work.

VIII-T6. I worry my skills will become outdated quickly because of AI.

VIII-T7. (Attention check) For this item, please select "Disagree."

Thank you for taking part in this study.

Appendix B

Answer Key and Scoring Rubric for the AI Proficiency Test

This appendix is for the researchers, the expert panel, and the raters. It is not shown to respondents.

Part A and Part B Answer Key

Item	Correct answer	Item	Correct answer
A-1	b	B-1	Prompt 2
A-2	c	B-2	Prompt 1
A-3	c	B-3	Prompt 2
A-4	a	B-4	Prompt 1
A-5	c	B-5	Prompt 2
A-6	d	B-6	Prompt 2
A-7	b		
A-8	b		
A-9	a		
A-10	d		

Note. Because the online form randomizes item order and option positions within parts, the key must be applied to the option text, not to its displayed position.

Part C Planted Errors (1 point each)

Passage	Error 1	Error 2	Error 3
C-1	Says "no added sugar" and then states 25 g of added sugar (contradiction).	"Clinically proven to cure fatigue" is an unsupported and misleading health claim.	"Number one coffee brand in the world" is an unsupported superlative claim.
C-2	Percentages add up to 110%; counts add up to 220 of 200 respondents.	Conclusion says most prefer in-store, but the data show online was highest.	Findings from a student sample are generalized to all Filipino consumers.
C-3	"Four-week" campaign, but dates March 1 to 14 cover two weeks.	1,000 clicks from 50,000 people is 2%, not 20%.	PHP 20,000 / 1,000 clicks is PHP 20 per click, not PHP 40.
C-4	Penetration pricing means a low entry price; the description given is skimming.	Two named competitors contradict "monopoly with a single dominant seller."	Ignoring competitor pricing contradicts the price-sensitive market it describes.

Part D Rubric (0 to 6 points per task)

Award one point for each criterion that is clearly met. Raters score independently and without seeing respondent identity, then record scores separately for reliability analysis. For the final score, the average of the two raters' scores is used (rounded to one decimal).

Criterion	Awarded when the response...
1. Clear goal	States the specific task and the output wanted (for example, a one-week plan with posts per day).
2. Audience and context	Names the business, target audience, and relevant context (for example, new shop, local market).
3. Constraints and format	Specifies length, tone, structure, or other limits for the output.
4. Verification plan	Describes concrete checks (confirm facts against the source, test claims, edit for accuracy and fit) before using the output.
5. Privacy and ethics	Avoids pasting confidential or personal data, or addresses originality, disclosure, or misleading claims.
6. Marketing judgment	Ties the work to a business objective and keeps a human decision in the process (for example, tailoring to local customers, approving final content).

Total Score

Total points = Part A (10) + Part B (6) + Part C (12) + Part D (12) = 40. AI Proficiency Score = (total points / 40) x 100.

Appendix C

Job Posting Codebook

Coding Sheet Variables

Variable	Coding rule
Posting ID, date, platform	Assign a unique ID; record the verified posting date and the platform or archive source.
Period	Pre (Dec. 1, 2021 to Nov. 29, 2022) or post (Nov. 1, 2025 to Oct. 31, 2026).
Role family	1 = Digital marketing and social media; 2 = Content and copywriting; 3 = Brand or product marketing; 4 = Market research and analytics; 5 = Advertising, campaign, or account management; 6 = General marketing officer, assistant, or associate.
Industry	1 = Retail or consumer goods; 2 = IT-BPM or technology; 3 = Hospitality and tourism; 4 = Real estate or construction; 5 = Banking, finance, or insurance; 6 = Education or health; 7 = Media or advertising agency; 8 = Other.
Location type	1 = Philippines on-site or hybrid; 2 = Philippines remote; 3 = Remote open to Filipino applicants (overseas employer).
Years of experience required	Enter the minimum number of years stated; 0 if none or if stated as open to fresh graduates.
Entry-level	1 if one year or less of experience is required or the posting is open to fresh graduates; otherwise 0.
Education requirement	1 if a bachelor's degree (or marketing-related degree) is stated; otherwise 0.
Portfolio required	1 if a portfolio or work samples are required or requested; otherwise 0.
AI skill present	1 if at least one competency in the AI skill categories is listed as required or preferred; otherwise 0.
AI skill required	1 if the AI competency is stated as required or in the main qualifications; 0 if only preferred or absent.
AI category	Mark every category that applies: tool proficiency; prompting; AI-assisted content creation; AI-assisted research and analytics; evaluation and responsible use; general AI familiarity.
Named AI tools	Record each AI tool named (for example, ChatGPT, Gemini, Copilot).
AI-related duties	Record any duties that describe new AI-related tasks (for example, building prompt workflows, reviewing AI-generated content).

Decision Rules

- Count a mention only when it refers to AI or generative AI in the sense of this study. Terms such as "AI" used in another sense, or purely traditional marketing automation with no AI reference, are not coded as AI skills.
- A keyword list (for example, "AI," "artificial intelligence," "generative AI," "ChatGPT," "prompt") is used to find possible mentions. The coder then reads the posting to decide whether the mention qualifies.

- If a posting lists an AI competency under both required and preferred sections, code it as required.
- If a posting is cross-listed on several platforms, keep the first dated instance only.
- When coders disagree, discuss and apply the rule above that fits; record unresolved cases and decide with a third member.

Appendix D

Expert Validation Form

Dear Expert Validator:

You are invited to review the research instruments for the study *The Impact of Artificial Intelligence on Employment Opportunities for Marketing Management Graduates*. Please rate each item for **relevance** to the research question it is meant to measure using the scale below, and give any comments on wording, clarity, or difficulty. For the AI proficiency test, please also comment on whether the correct answers are accurate and whether the items are suitable for Marketing Management graduates.

Rating	Meaning
4	Highly relevant
3	Quite relevant (needs minor revision)
2	Somewhat relevant (needs major revision)
1	Not relevant

Rating Sheet

Part and item	Related research question	Rating (1 to 4)	Comments and suggested revision
Part I (all items)	Respondent description; eligibility		
Part II (II-1 to II-8)	RQ2, RQ3 (control variables)		
Part III (III-1 to III-11)	RQ2, RQ3 (employment outcomes)		
Part IV (IV-1 to IV-8)	RQ4a		
Part V-A (A-1 to A-10)	RQ2 (AI proficiency)		
Part V-B (B-1 to B-6)	RQ2 (AI proficiency)		
Part V-C (C-1 to C-4)	RQ2 (AI proficiency)		
Part V-D (D-1, D-2) and rubric	RQ2 (AI proficiency)		
Part VI (VI-1 to VI-6)	RQ2a		
Part VII-A (tasks)	RQ3a		
Part VII-B (VII-B1 to VII-B8)	RQ3b		
Part VIII (opportunities, threats)	RQ4b		
Job posting codebook	RQ1, RQ3b		

Note. For a full review, rate each individual item. This sheet lists groups of items to show the layout; the final form will list every item separately.

Name of expert: _____ Position and institution: _____

Signature: _____ Date: _____